SLR coding practice

Your name

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```r
library(tidyverse)
library(openintro)
library(broom)
```

1. Using the `possum` data from `openintro`, fit a linear regression for the length of heads for a possum using their tail lengths as a predictor. Store the model output as a variable called `possum_lm` and display information about the coefficients in *tidy* form using the appropriate function.

2. Create a residual plot of the fitted model using the `augment()` function and `ggplot()`. Try adding a horizontal dashed line at 0!

3. Suppose we found a tail of a possum on the ground! We measure the tail to be 36.1 cm. Being as reproducible as possible, obtain the estimated head length of this squirrel by doing the following:

- Create variable called `x_pred` that stores the value of the explanatory variable we'd like to obtain a prediction for.
- Obtain the estimated coefficients from the tidy output by `pull()`-ing the relevant column. Store them as a variable called `possum_coeffs`.
- Indexing `coeffs` (i.e. using bracket notation) and using `x`, obtain the estimated head length and store this as a value called `head_l_pred`.

```r
# create x_pred

# create possum_coeffs

# create head_l_pred
```

Report your answer using in-line code.

Answer: